Raphaël ARAKELIAN

Formal Methods for Distributed Systems

Apprentice Research Engineer at CEA-List
Master's student at Université Paris-Saclay

raphael.arakelian@universite-paris-saclay.fr
linkedin.com/in/raphaël-arakelian Orsay, France

PROFILE

Specialised in the **formal specification and verification** of computing systems, applied to **fault-tolerant distributed systems** (including Byzantine faults). Experience in **deductive verification**, compositional assume-guarantee reasoning, specification-based testing and formal semantics, aimed at **safety- and security-critical systems**. Available from September 2028.

EXPERIENCE

Research Engineer in Formal Methods for Trustworthy Distributed Computing — Sep 2026 – Sep 2028
CEA-List – Commissariat à l'énergie atomique et aux énergies alternatives, Nano-Innov — Palaiseau, France

Research and development within the team behind **Ledgera**, an open-source distributed ledger that certifies, through cryptographic proofs, computations shared between parties who do not trust each other, in a Byzantine setting. Its applications include trustworthy collaborative AI. I am in charge of the **formal verification** side of the project, from the formal specification of the system's behaviour to the tests derived from it and run against its **Rust** implementation, and contribute to the design of a formally specified language for verifiable computations. A complete immersion in research and development, from formal modelling to implementation and scientific writing. Supervised by Álvaro García Pérez.

Research Intern in Formal Verification of Smart Contracts — May 2026 – Jul 2026
LIP6 – Laboratoire d'Informatique de Paris 6, Sorbonne Université — Paris, France

Research on making smart contracts provably safe, in collaboration with Dowsers, a blockchain security company. I extended **VeriSol**, Microsoft Research's deductive verifier for smart contracts built on **Boogie**, so that it can handle contracts written in several languages (Solidity, Cairo, Rust), with a working prototype integrated into the tool. I formalised the translation from source code to the verification language and designed **compositional (assume-guarantee) reasoning** rules, so that each contract can be verified independently of the others. Supervised by Souheib Baarir and Nicolas Nalpon.

Research Intern in Bitcoin Ransomware Tracing with Machine Learning — Jan 2026 – Apr 2026
LMF – Laboratoire Méthodes Formelles, ENS Paris-Saclay — Saclay, France

I built a tool tracing ransomware payments through the **Bitcoin transaction graph** down to cash-out addresses, combining address clustering heuristics with **machine learning** classification of laundering patterns (mixers, peel chains) and of the entities encountered. Supervised by Sylvain Conchon.

EDUCATION

Université Paris-Saclay — **Orsay, France**

Master's degree in Computer Science for Data Science — 2026 – 2028
Computer science programme focused on large-scale data and distributed systems, combining machine learning, distributed data processing and cloud computing with dedicated courses on **blockchain and smart contracts**, and on **responsible AI**. Followed as an apprenticeship, alternating between the university and CEA-List.

Bachelor's Double Degree in Mathematics and Computer Science, Magistère d'Informatique — 2023 – 2026
Selective double degree in mathematics and computer science, completed with the **Magistère d'Informatique**, Paris-Saclay's research-oriented programme of excellence. Strong foundations in mathematics and theoretical computer science, advanced courses in **compilation**, lambda calculus (ENS Paris-Saclay), **distributed algorithms** and software safety and security, and early immersion in research laboratories.

Lycée Maurice Genevoix — **Montrouge, France**

French Baccalauréat in Mathematics and Computer Science — 2021 – 2023

SKILLS

Formal Verification	Deductive verification (Boogie, VeriSol), pre/postconditions and invariants, compositional assume-guarantee reasoning, formal semantics (BNF grammars, inference rules), specification-based and property-based testing, lambda calculus
Distributed Systems	Byzantine fault tolerance and consensus, distributed ledgers and smart contracts (Solidity, Cairo, ink!), verifiable distributed computation, concurrent and parallel programming
Data & AI	Machine learning and deep learning, data mining, optimisation, natural language processing, responsible and trustworthy AI, graph analysis
Programming	Python, OCaml, C, C#, Rust, Haskell, C++, Java, SQL, MIPS assembly (Mini Go compiler)
Tools	Rocq (Coq) and Lean (basics), QuickCheck, Git, Linux, LaTeX
Languages	French (native), English (C1, TOEIC 960/990), German (upper intermediate)